

Induction: Gauss S 40

$$\begin{array}{l} \textcircled{1} + 2 + \dots + 100 \quad = S \\ \textcircled{100} + 99 + \dots + 1 \quad = S \end{array}$$

$$101 + 101 \dots + 101 = \frac{2S}{2} = 101 \cdot 100$$

101

$$S = \frac{101 \cdot 100}{2}$$

WTP: $\forall n \in \mathbb{Z}_{>0}$

$$1 + 2 + \dots + n = \sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$n=1$

$$1 = \frac{1(1+1)}{2} = 1 \quad P(n)$$

$n=2$

$$1+2 = \frac{1+2}{2} + 2 = 2 \left(\frac{1}{2} + 1 \right) = 2 \left(\frac{3}{2} \right)$$

$n=3$

$$1+2+3 = \frac{2(3)}{2} + 3 = 3 \left(\frac{2}{2} + 1 \right) = 3 \left(\frac{4}{2} \right)$$

$$1+2+3+4 = \frac{3 \cdot 4}{2} + 4 = 4 \left(\frac{3}{2} + 1 \right) = 4 \left(\frac{5}{2} \right)$$

Proof, Proceed by induction. The statement is
already true for $n=1$ b/c $1 = \frac{1(2)}{2} = 1$.

^{"Assume $P(n)$ "}
Assume that we already know the statement
for n . IE assume that $\leftarrow$ this is $P(n)$.

$(1 + 2 + \dots + n = \frac{n(n+1)}{2})$
Adding $n+1$ to both sides gives

$$1 + 2 + \dots + n + (n+1) = \frac{n(n+1)}{2} + (n+1).$$

The LHS of this is the LHS of
what we want to prove. The RHS is

$$(n+1) \left(\frac{1}{2} + 1 \right) = (n+1) \left(\frac{n+2}{2} \right) = (n+1) \frac{(n+1)!}{2}$$

We conclude that $\forall n \in \mathbb{Z}_{\geq 0}$,

$$1+2+\dots+n \geq \frac{n(n+1)}{2}. \quad \square$$

to this is
a proof that
 $P(n) \Rightarrow P(n+1)$.

Let $P(n)$ be a statement which depends on some integer (usually pos) n .

(Ex: $P(n) = "1 + 2 + \dots + n = \frac{n(n+1)}{2}"$)

Goal: Prove $P(n) \forall n \in \mathbb{Z}_{\geq 0}$.

Step 1: Prove $P(1)$ "Base Case"

Step 2: Prove " $P(n) \Rightarrow P(n+1)$ " "Inductive Step"

"Induction" = $P(1) \wedge \underbrace{P(n) \Rightarrow P(n+1)}_{\forall n \in \mathbb{Z}_{\geq 0}, P(n)} \Rightarrow$

$P(1), P(2), P(3), P(4), \dots, P(a), P(a+1), \dots$

Warning: $P(n) \neq \boxed{\frac{n(n+1)}{2}}$ ← not "t or F"

$$P(n) = 1+2+\dots+n = \frac{n(n+1)}{2}$$

Note: n is just a variable.

$$P(1) \wedge P(n) \Rightarrow P(n+1)$$

$$P(a) \Rightarrow P(a+1)$$

$$P(a-1) \Rightarrow P(a)$$

"prove the following case!"

Voraussetzung: $P(0) \wedge P(n) \Rightarrow P(n+1)$

$$P(a) \wedge P(n) \Rightarrow P(n+1)$$

$$P(a) \wedge P(n) \Rightarrow P(n+a)$$

$$\Rightarrow \forall n \in \mathbb{N}_{\geq 0}, P(n)$$

$$P(-1) \wedge P(n) \Rightarrow P(n-1) \quad |$$

$$\forall n \in \mathbb{Z}_{\leq 0}, P(n)$$

$P(a), P(n), P(-)$

Define: a sequence $a_1, a_2, \dots, a_n, \dots$

$$a_1 = 2 = 2^1$$

"recursive defn"

$$a_n = 2 \cdot a_{n-1}$$

$$a_n = 2 \cdot a_{n-1}$$

$$a_{n+1} = 2 \cdot a_n$$

$$a_2 = 2 \cdot a_1 = 2 \cdot 2 = 4 = 2^2$$

$$a_3 = 2 \cdot a_2 = 2 \cdot 4 = 8 = 2^3$$

$$a_4 = 2 \cdot a_3 = 2 \cdot 8 = 16 = 2^4$$

Claim: $\forall n \in \mathbb{Z}_{70}, \boxed{a_n = 2^n}$

$$p(n) = "a_n = 2^n"$$

Proof: Proceed by induction.

Base case: $P(1) \models "a_1 = 2"$, i.e., $2 = 2$.

($P(n) \Rightarrow P(n+1)$) This is true.

Inductive step: Assume $P(n)$, i.e., $a_n = 2^n$.

(wtp $P(n+1)$, i.e., $a_{n+1} = 2^{n+1}$) Then $a_{n+1} = 2 \cdot a_n$

by the defn. of a_n . Then $a_{n+1} = 2 \cdot 2^n = 2^{n+1}$.

Thus $P(n+1)$ is true. $\square$

Define: $a_1 = 0$

$$a_n = \sqrt{3 + 2a_{n-1}}$$

$$a_1 = 0$$

$$a_2 = \sqrt{3 + 2 \cdot 0} = \sqrt{3}$$

$$a_3 = \sqrt{3 + 2a_2} = \sqrt{3 + 2\sqrt{3}}$$

$$a_4 = \sqrt{3 + 2a_3} = \sqrt{3 + 2\sqrt{3 + 2\sqrt{3}}}$$

Prop: $\forall n \in \mathbb{Z}_{>0}$

$$a_n < 3$$

$$P(n) = "a_n < 3"$$

Proof: Proceed by induction.

Base case: $P(1)$ is " $a_1 < 3$ ", i.e., " $0 < 3$ ".

This is true.

I.S. Assume $P(n)$, i.e., " $a_n < 3$ ".

[WTP: $P(n+1)$, i.e., " $a_{n+1} < 3$ "] Then, by defn,

$a_{n+1} = \sqrt{3 + 2a_n}$. Since $a_n < 3$,

$$a_{n+1} = \sqrt{3 + 2a_n} < \sqrt{3 + 2 \cdot 3} = \sqrt{3+6} \\ = \sqrt{9} = 3$$

Thus $a_{n+1} < 3$. $\square$

Claim: " $1 + 2 + 4 + 8 + \dots + 2^{n-1} = 2^n - 1$ "

Proof: Proceed by induction. $\forall n \in \mathbb{Z}_{>0}$

BC: $P(1)$ is " $1 = 2 - 1$ ". This is true

IS: Assume $P(i)$, i.e., $1 + 2 + \dots + 2^{i-1} = 2^i - 1$.

(wrt $P(i+1)$, i.e., $1 + 2 + \dots + 2^{i+1} = 2^{i+1} - 1$)

$$1 + 2 + \dots + 2^{i-1} + 2^i = 2^{i+1} - 1$$

Adding a^i to each side gives

$$1 + a + \dots + a^{i-1} + a^i = a^{i-1} + a^i.$$

The LHS is the LHS of $P(i+1)$.

The RHS is $a - a^{i-1} = a^i - 1$.

This is the RHS of $P(i+1)$. $\square$

Claim: $\forall n \in \mathbb{Z}_{\geq 0}, \underbrace{3 \mid 4^n - 1}_{P(n)}.$

$P(n)$

Proof: Proceed by induction.

Bx: $P(0)$ is " $3 \mid 4^0 - 1$ ", i.e., " $3 \mid 0$ "

which is true.

IS: Assume $P(n)$, i.e., $3|4^n - 1$.

(wtp $3|4^{n+1} - 1$) Then $\exists m \in \mathbb{Z}$ s.t.

$$4^n - 1 = 3m. \text{ Then } 4^n = 3m + 1.$$

$$\text{Then } 4^{n+1} - 1 = 4^n \cdot 4 - 1 = (3m + 1) \cdot 4 - 1$$

$$= 12m + 4 - 1 = 12m + 3 = 3(4m + 1).$$

thus

$$3/4^{n+1} - 1$$



